Summary

This assignment uses the Hough Transform to detect circles and lines in images. The Hough transform was applied to two images:

1. Line detection using the polar Hough transform in the road image for line detection to detect the longest line in the road image using the polar Hough transform variables (ρ, θ).
2. Circle detection using a circle-centre accumulator for a given radius in an image containing coins but also diagonal lines, but we are focusing on the circles using three different radii between 30-40 as required in tutorial and the radii chosen was (R = 30,31,32).

For both questions I transformed the image into grayscale, then did Canny edge detection, then parameter space quantization, then I accumulated votes, and then I extracted peaks (local maxima), and finally I overlaid the detected shapes on the original image to show the output of all detected lines.

Basic Understanding of the Hough Transform

Hough Transform is an algorithm based on votes that converts edge points in an image into a shape parameter space. Each edge point votes for all possible shapes (lines or circles) that pass through that point. Peaks in the accumulator correspond to good shape candidates.

Discussion

Method Used for Question 1

1. I applied on the road image (test1) conversion to grayscale and then used Canny edge detector to generate a binary edge map.
2. I did ROI constraint, as in the initial testing, the presence of strong edges around the horizon and background could produce strong peaks in the Hough space and thus non-road detections. To ensure that the detection is centered on the road area, a polygonal region of interest (ROI) mask was applied to the edge map. This ensured that only pixels within the region of interest (ROI) casted their votes in the Hough space, so apply the ROI mask solved this issue that will be discussed later in the report, as shown in figure 1.

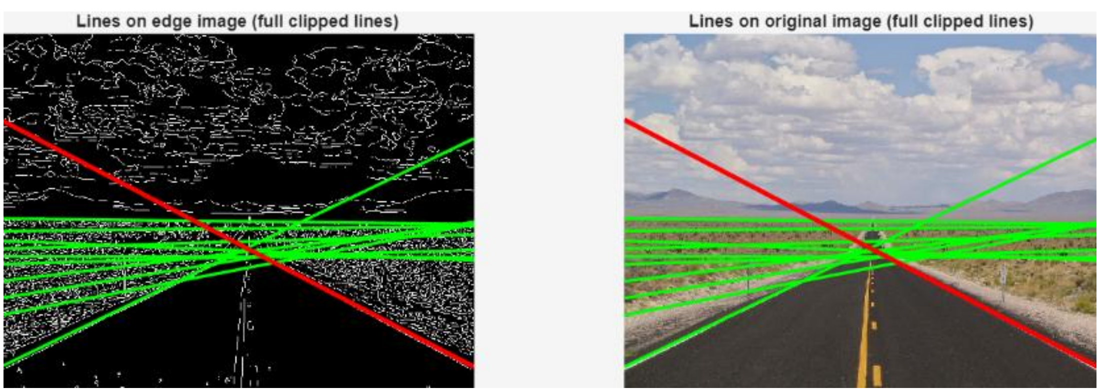

Figure 1: detected lines without ROI

3. Lines are represented in polar form to avoid the vertical-line slope issue:

$$\rho = x\cos(\theta) + y\sin(\theta)$$

Where x: row index, y: column index, ρ is the perpendicular distance from the origin and θ is the angle of the normal as shown in figure 2.

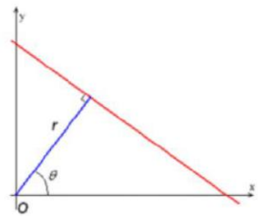

Figure 2: polar Hough Transform equation

The parameter spaced was added as following:

$$\theta\ was\ limited\ to\ reduce\ shallow\ angle\ lines$$

ρ spans the image diagonal range with step $\Delta\ \rho = 1$ pixel

4. The formula used for voting was acc(ρi,θj)=acc(ρi,θj)+1
5. Then we detected the peaks (local maxima) in $A(\rho,\theta)$ using non-maximum suppression, and the top peaks above the threshold was taken as candidate lines.
6. Finally, the infinite line is clipped to the edges of the image to show each potential line as a visible line segment. Each clipped line segment's Euclidean length is determined, and the longest line segment is selected as the output. Additionally, as a quantitative indicator of line strength, the number of supporting edge pixels near the line is determined using a distance tolerance.

Results for Question 1

Output results from question one line detection on the road image:

Q1 Result (Longest Line)
rho = -120.00
theta = 2.6445 rad
p1 = (row=137, col=1)
p2 = (row=571, col=800)
length (pixels) = 909.00
support = 430

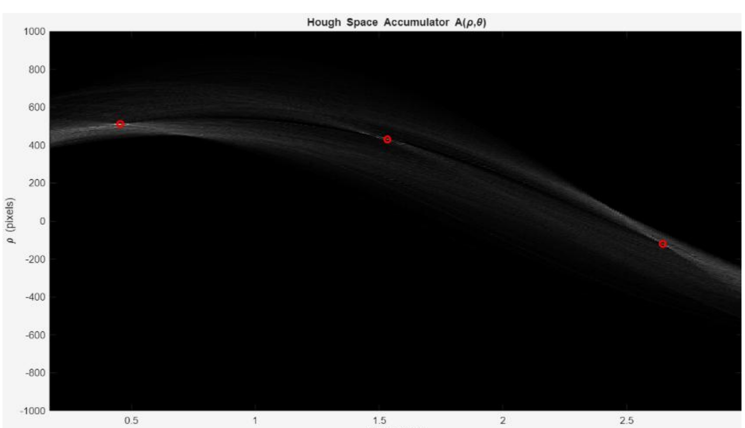

Figure 3: Hough Transform of the road image (test1)

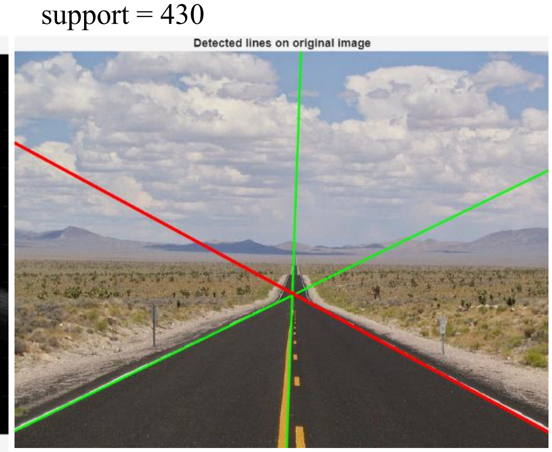

Figure 4: Output detected lines with applied parameters to only detect road

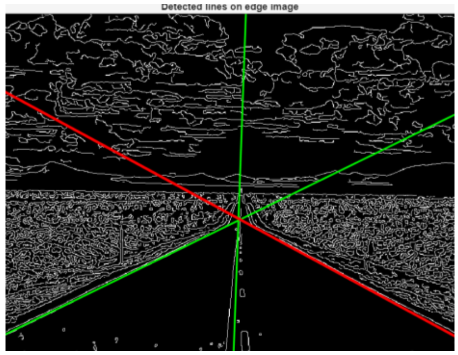

Figure 5: Canny edge detected output

Analysis of the Results of Question 1

The Hough transform accumulator reveals prominent peaks reflecting the dominant line features in the image. In road images, edge information from lane markings and road boundaries casts consistent votes, resulting in prominent peaks in the parameter space. Without the constraints, strong long edges in the background, such as boundaries of the horizon region, could potentially cast dominant votes and lead to selection based on a "longest line" criterion which will be the horizon. Using an ROI restricts votes to come only from edges within the road region, and the selection of the longest line matches the objective of road detection for image processing, as of the understanding of the tutorial that we need to detect diagonal lines and also to be applied into image processing applications in real world, and detecting the road will be the main objective in real world application. The longest line selected has strong support (430 edge pixels), reflecting that a large number of edge pixels lie close to the detected line.

Method Used for Question 2

1. Converting image of the coins (test 2) into greyscale, then applying Canny edge detector on the image
2. For circle detection with a fixed radius R, the unknowns are the circle center (a,b). The parameter space is (a,b), where: a is the center column, b is the center row.
3. Then voting where For each edge pixel (x,y) where x is column and y is row, and possible centers are: $a = x - R\cos(\theta)$, $b = y\ R \sin(\theta)$. And then we update the accumulator center by: $H(b,a)=H(b,a)+1$. By this method a 2D accumulator image is created where bright peaks correspond to likely circle centers.
4. Then peaks are extracted (local maxima) are extracted using non-maximum suppression, producing center of circles candidates.
5. Then we try three different radii (30,31,32) on the image, and I chose these values, because the coins are nearly identical in size and all are close to these values.

Results for Question 2

Output of the coins image (test2):

Q2 Calculated Radii
R = 30, centers found = 7
R = 31, centers found = 4
R = 32, centers found = 7

Q2 Result (Circle centers)
Used radii: [30 31 32]
Total circles detected (all radii) = 18

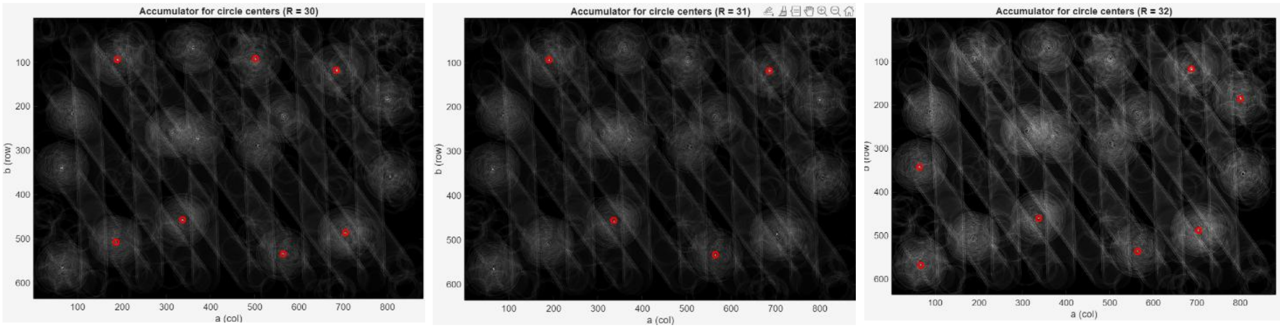

Figure 6: detected circle centers using different radii (30,31,32)

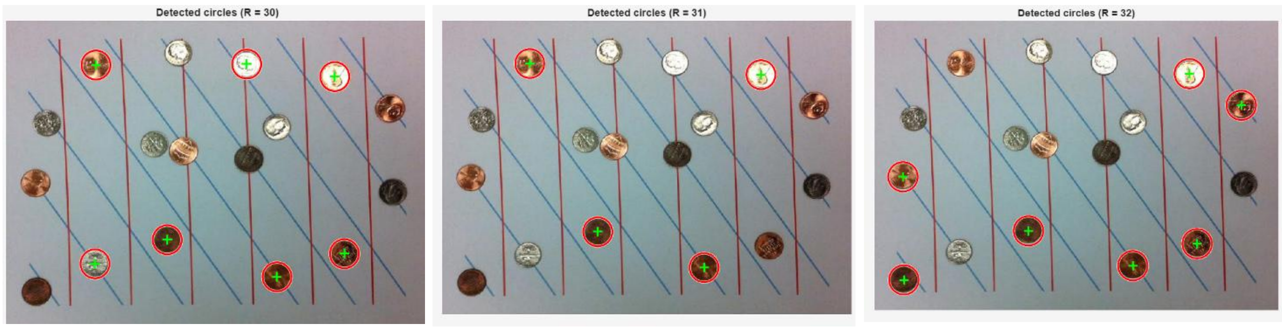

Figure 6: detected coins using different radii (30,31,32)

Analysis of the Results of Question 2

With a radius R, boundary pixels cast votes for possible circle centers, and hence, actual circle boundaries cause constructive voting around the actual center. This causes high peaks in the 2D accumulator array. In the coin image (test2), the physical sizes of most coins are similar, and hence, the projected pixel radius is similar. Thus, the use of adjacent radii values [30,31,32] causes the detection of many similar coins in different runs because the circle model continues to fit the same edge pixels and generate peaks around the same centers, though with varying intensities depending on the value of R. Because the coins are of similar sizes, radii values of 30 to 32 pixels will generate valid peaks and possibly detect partially hidden coins or generate additional peaks from strong background edges and overlapping objects. A higher peak threshold will reduce false positives but may fail to detect weak circles, while a lower threshold will increase detection capabilities but generate additional false positives. And I chose these reasonable values to detect the coins which are very similar in radius and satisfy the assignment requirement of detecting circles for at least three different radii. This behavior is shown in the results, where $R = 30$, $R = 31$, and $R = 32$ detected 7, 4, and 7 centers respectively, for a total of 18 detected centers across the three radii.

Conclusion

This assignment required applying the Hough Transform for line and circle detection (as said in the tutorial video, the circle detection in easier than the line detection). For road image (test1), the longest line was successfully detected and displayed together with the candidate lines and the peaks of the Hough accumulator. A road region ROI was applied to suppress false positives due to the horizon. Circle detection was applied using center voting for fixed radii, and circles were successfully detected for three different radii R =30,31,32. The results validated how peaks in Hough parameter space relate to geometric structures in image space.

For coins image (test2), circle detection was applied using center voting for a fixed radius R. Following the tutorial instructions to test radii in the range (30-40). R=(30,31,32) were chosen since they are very close to the projected pixel radius of most coins in the image. The approach successfully detected circles at these radii and showed how high peaks in Hough parameter space relate to significant geometric structures (lines and circles) in image space.